Integration

Week 5B

Overview

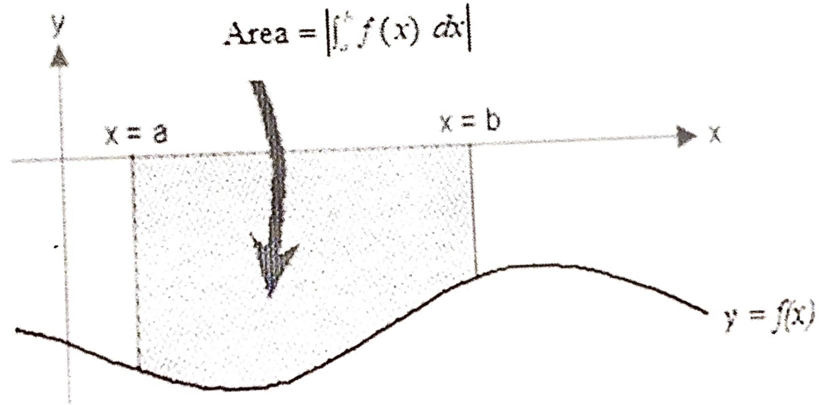
Definite Integrals and Area

Key Ideas

- *Area below the x-axis*
  - *Subdividing Regions before Integrating*
  - *Symmetrical Graphs*
- *Area between two Curves*

Key Idea – Area below the x-axis

When the region lies below the x-axis, the definite integral gives a negative value. To find the area, take the absolute value of the integral or place a negative sign in front of the integral, to ensures the area is positive.



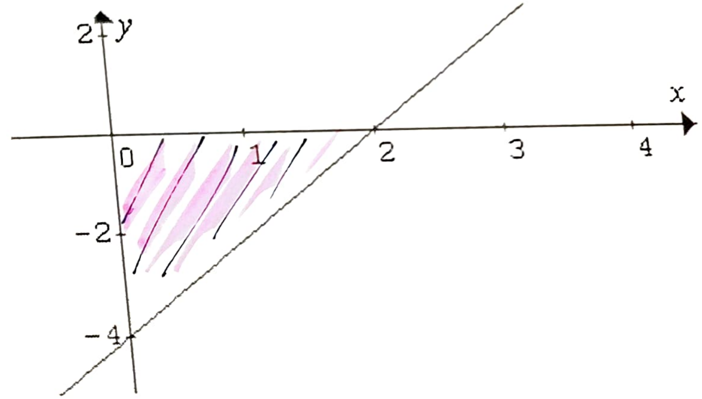
$|-5| = 5$

$-(-5) = 5$

$$A = \left|\int_a^b f(x)\,dx\right| \quad \textbf{or} \quad A = -\int_a^b f(x)\,dx$$

Example:

Calculate the area enclosed by the x and y – axes and the graph $y = 2x - 4$.



$$A = -\int_a^b f(x)\,dx$$

$$= -\int_0^2 (2x-4)\,dx \quad \text{OR} \quad \left| \int_0^2 (2x-4)\,dx \right|$$

$$= -\left[x^2 - 4x\right]_0^2$$

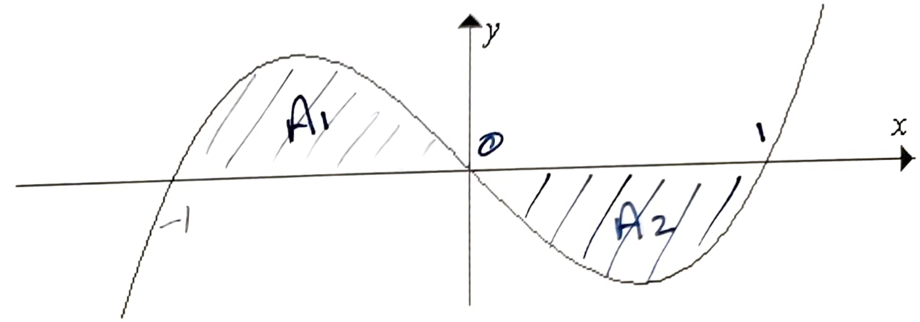
$$= -\left(\left(2^2 - 4\times 2\right) - \left(0^2 - 4\times 0\right)\right)$$

$$= -\left((4) - (0)\right)$$

$$= 4 \text{ units}^2$$

Exercise:

Find the area enclosed between the x-axis and the curve $y = x^3 - x$ from $x = 0$ to $x = 1$.



$$A = \left| \int_0^1 (x^3 - x^1)\,dx \right|$$

$$= \left| \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 \right|$$

$$= \left| \left[\left(\frac{1^4}{4} - \frac{1^2}{2} \right) - \left(\frac{0^4}{4} - \frac{0^2}{2} \right) \right] \right|$$

$$= \left| \left(\frac{1}{4} - \frac{1}{2} \right) \right|$$

$$= \left| -\frac{1}{4} \right| = \frac{1}{4}\,u^2$$

$A = A_1 + A_2$

$= \frac{1}{4} + \frac{1}{4}$ or $2 \times \frac{1}{4}$

$= \frac{1}{2}\,u^2$ $= \frac{1}{2}u^2$

$$A = \int_{-1}^{0} (x^3 - x)\,dx + \left| \int_0^1 (x^3 - x)\,dx \right|$$

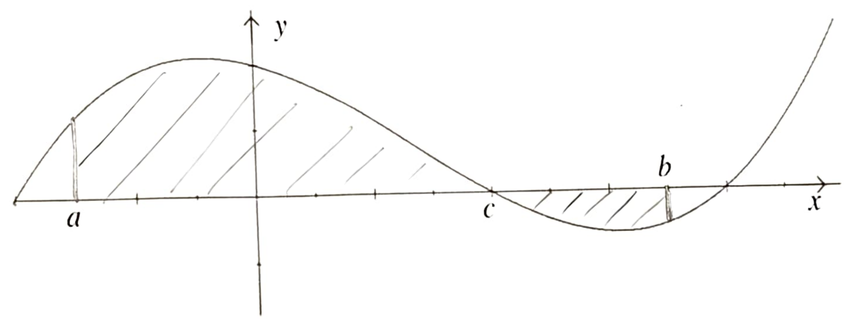
$$= \frac{1}{2}u^2$$

$$\int_{-1}^{1} (x^3 - x)\,dx =$$

Subdividing Regions before Integrating

If the region between the curve and the x-axis contains areas both **above** and **below** the x-axis, the region must be divided into separate parts. The area of each part is then calculated separately and added together to find the total area.

Example: Find the area between the curve the x-axis from a to b.

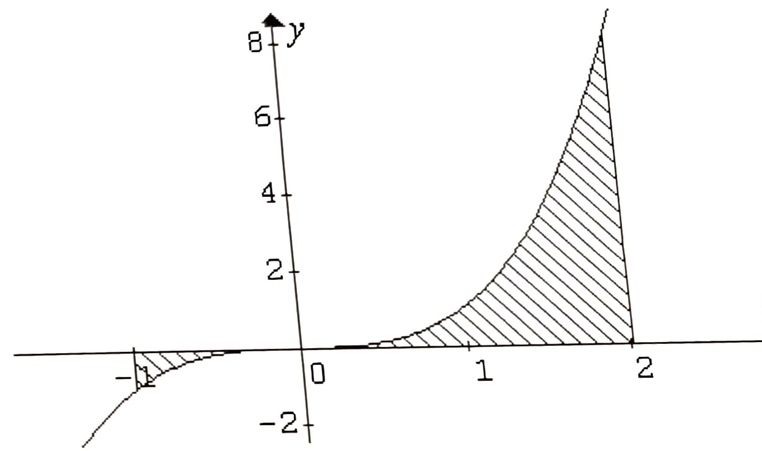


The area between the curve and the x-axis from a to b is **NOT equal to** $A = \int_a^b f(x)\,dx$.

Instead, it is $A = \int_a^c f(x)\,dx + \left| \int_c^b f(x)\,dx \right|$.

Example:

Find the area enclosed between the x-axis and the curve $y = x^3$ from $x = -1$ to $x = 2$.

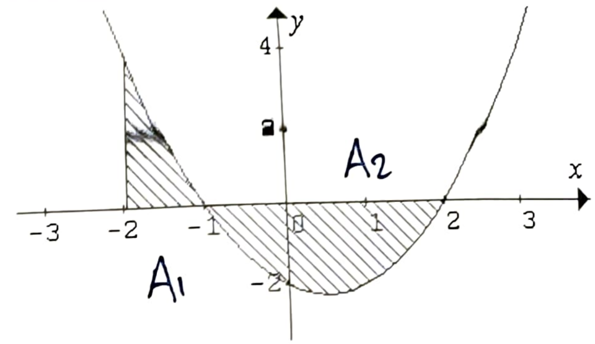


$$\begin{aligned} \text{Area} &= \text{Area } 1 + \text{Area } 2 \\ &= -\int_{-1}^{0} x^3\,dx + \int_{0}^{2} x^3\,dx \\ &= \left[-\frac{x^4}{4}\right]_{-1}^{0} + \left[\frac{x^4}{4}\right]_{0}^{2} \\ &= \left[-0-(-\frac{1}{4})\right] + \left[\frac{16}{4}-0\right] \\ &= \frac{1}{4}+4 \\ &= 4\tfrac{1}{4} \text{ units}^2 \end{aligned}$$

Exercise:

The graph of $y = x^2 - x - 2$ is drawn below.

Write an integral which can be used to calculate the shaded area.



. And evaluate

$$A = A_1 + A_2$$

$$= \int_{-2}^{-1} (x^2 - x - 2)\,dx + \left| \int_{-1}^{2} (x^2 - x - 2)\,dx \right|$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x\right]_{-2}^{-1} + \left| \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x\right]_{-1}^{2} \right|$$

$$= \left[\left(\frac{(-1)^3}{3} - \frac{(-1)^2}{2} - 2\times(-1)\right) - \left(\frac{(-2)^3}{3} - \frac{(-2)^2}{2} - 2\times(-2)\right)\right]$$

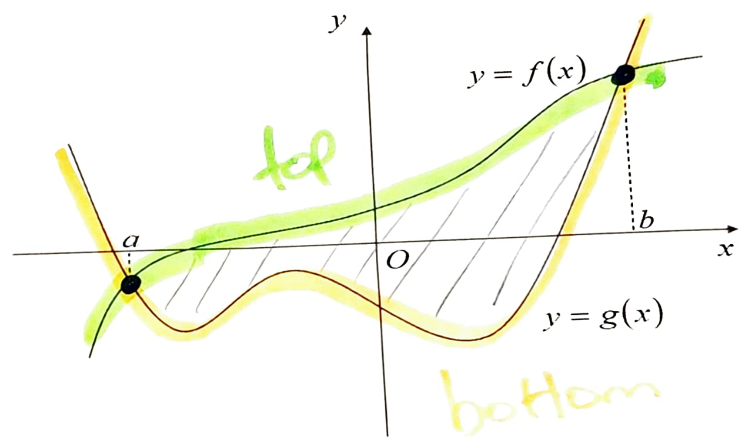
$$+ \left| \left[\left(\frac{2^3}{3} - \frac{2^2}{2} - 2\times 2\right) - \left(\frac{(-1)^3}{3} - \frac{(-1)^2}{2} - 2\times(\right.\right.\right.$$

$$= \frac{9}{2} + \frac{11}{6}$$

$$= \frac{19}{3}\ u^2$$

Key Idea - Area between Two Curves

Sometimes we want to find, not the area between a curve and the x-axis, but the area enclosed between two curves. If $f(x)$ and $g(x)$ are continuous with $f(x) \geq g(x)$ for x in $[a,b]$.



Then the area A of the region enclosed by the upper curves $y = f(x)$ and the lower curve $y = g(x)$ and by the ordinates $x = a$ and $x = b$ is given by

$$\boldsymbol{A = \int_a^b \left(f(x) - g(x)\right) dx}$$

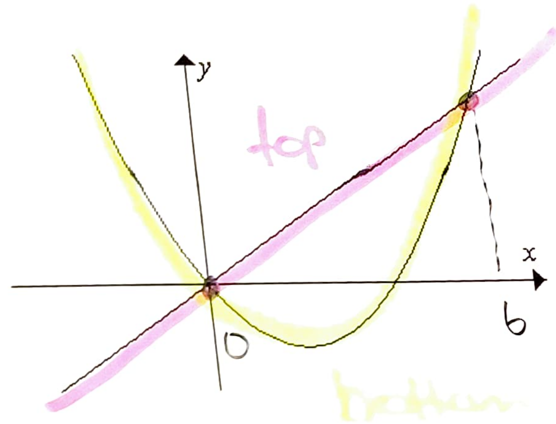
Note: As long as $f(x) \geq g(x)$ for x in $[a,b]$, we do not have to worry about whether the curves cross the x-axis.

So, to find the area enclosed between two curves, we must:

1. Find where the curves intersect.
2. Find which is the upper and lower curve in the region you are interested in.
3. Integrate the function (upper curve – lower curve) between the appropriate limits.

***Example*:**

Write the integral which can be used to find the area of the region bounded by the curves $y = 2x$ and $y = x^2 - 4x$.



First, we need to find where the curve intersects:
To find where the curves intersect, it is necessary to simultaneously solve the equations: $y = 2x$ and $y = x^2 - 4x$.

$$x^2 - 4x = 2x$$

$$x^2 - 6x = 0$$

$$x(x-6) = 0$$

The curves intersect at $x = 0, 6$

Find the upper and lower curves:
The upper curve is $y = 2x$ (straight line) and the lower curve is $y = x^2 - 4x$ (parabola).

Integrate the function (upper curve − lower curve) between the appropriate limits:

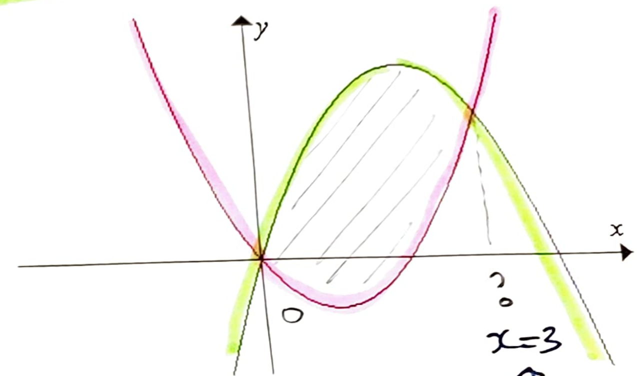
$$A = \int_0^6 2x - (x^2 - 4x)\,dx = \int_0^6 6x - x^2 \ dx$$

$2x - x^2 + 4x$

[Evaluation of this integral will give an area of 36 u²]

Exercise:

Find the area of the region bounded by the two parabolas $y = 4x - x^2$ and $y = x^2 - 2x$.



1. Where the curves intersect:

$$4x - x^2 = x^2 - 2x$$
$$4x + 2x = x^2 + x^2$$
$$2x^2 = 6x$$
$$2x^2 - 6x = 0$$
$$2x(x-3) = 0$$
$$2x = 0 \quad \& \quad x - 3 = 0$$
$$x = 0 \qquad\qquad x = 3$$

$$A = \int_0^3 (\underbrace{4x - x^2}_{\text{top}}) - (\underbrace{x^2 - 2x}_{\text{bottom}})\, dx$$
$$= \int_0^3 (4x - x^2 - x^2 + 2x)\, dx$$
$$= \int_0^3 (6x - 2x^2)\, dx$$

$$\left[\frac{6x^2}{2} - \frac{2x^3}{3}\right]_0^3 = \left[3x^2 - \frac{2x^3}{3}\right]_0^3$$
$$\left[\left(3 \times 3^2 - \frac{2 \times 3^3}{3}\right) - (0)\right]$$
$$= 27 - 18$$
$$= 9\ u^2$$

Key Idea - Integration of Exponential Functions

Let's remind ourselves about the result of which states that integration and differentiation are inverse processes.

$$\frac{d}{dx}F(x) = f(x) \Leftrightarrow \int f(x)\,dx = F(x) + c$$

So,

$$\text{If } \frac{d}{dx}e^x = e^x \text{ and } \frac{d}{dx}e^{f(x)} = f'(x)e^{f(x)}$$

$y = e^{3x^2}$ $y' = 6xe^{3x^2}$

$\int 6xe^{3x^2}\,dx = e^{3x^2} + c$

It follows that,

$$\int e^x\,dx = e^x + c \text{ and } \int f'(x)e^{f(x)}\,dx = e^{f(x)} + c$$

Example: Integrate the following:

(a) $\int$ 2e[redacted] dx = e[redacted] + c

$\int 2e^{2x} = e^{2x} + c$

(b) $\int$ 3x[redacted]e[redacted] dx = e[redacted] + c

$\int 3x^2 e^{x^3} = e^{x^3} + c$

Discussion Question:

Does the integral of $\displaystyle\int 5e^{5x}\,dx$ fit the formula $\displaystyle\int f'(x)e^{f(x)}\,dx$? Yes

Does the integral of $\displaystyle\frac{1}{5}\int 5e^{5x}\,dx$ fit the formula $\displaystyle\int f'(x)e^{f(x)}\,dx$? No

What do we do to match the formula?

$$\frac{1}{5}\int 5e^{5x}\,dx = \frac{1}{5}e^{5x} + c$$

Example: Integrate the following:

(a) $$\int e^{2x}\,dx = \frac{1}{2}\int 2e^{2x}\,dx = \frac{1}{2}e^{2x} + c$$

(b) $$\int xe^{3x^2-1}\,dx = \frac{1}{6}\int 6xe^{3x^2-1}\,dx = \frac{1}{6}e^{3x^2-1} + c$$

(c) $$\int 5e^{2x}\,dx = 5\int e^{2x}\,dx = 5\times\frac{1}{2}\int 2e^{2x}\,dx = \frac{5}{2}e^{2x} + c$$

(d) $$\int e^{-3x+5}\,dx = -\frac{1}{3}\int -3xe^{-3x+5}\,dx = -\frac{1}{3}e^{-3x+5} + c$$

(e) $$\int e^{\frac{x}{3}}\,dx = 3\int \frac{1}{3}e^{\frac{x}{3}}\,dx = 3e^{\frac{x}{3}} + c$$

(f) $$\int 5xe^{-3x^2+1}\,dx = 5\int xe^{-3x^2+1}\,dx = 5\times\frac{-1}{6}\int -6xe^{-3x+5}\,dx = -\frac{5}{6}e^{-3x^2+1} + c$$

$$\int f'(x)e^{f(x)}\,dx = e^{f(x)} + c$$

Exercise:

Integrate the following:

(a) $\int 3e^{3x-2}\,dx = e^{3x-2} + c$

$f(x) = 3x-2$

$f'(x) = 3$

$f(x) = 2x^3-2$

$f'(x) = 6x^2$

(b) $\int \left(6x^2e^{2x^3-2}+3\right)dx = \int 6x^2e^{2x^3-2}\,dx + \int 3\,dx$

$= e^{2x^3-2} + 3x + c$

Exercise:

Integrate the following:

(a) $\int e^{-3x} + 2e^{2x-3}\,dx$

$f(x) = -3x$

$f'(x) = -3$

$= -\frac{1}{3}\int -3e^{-3x}\,dx + \int 2e^{2x-3}\,dx$ ✓

$= -\frac{1}{3}e^{-3x} + e^{2x-3} + c$

$a^{-n} = \frac{1}{a^n}$

(b) $\int \frac{2x}{e^{x^2}}\,dx$

$f(x) = -x^2$

$f'(x) = -2x$

$= -1\int -1\, 2x e^{-x^2}\,dx$

$= -e^{-x^2} + c$

$= \frac{-1}{e^{x^2}} + c$

Exercise:

Integrate the following:

(a) $\int \left(e^{\frac{x}{2}} - x^2\right) dx$

$f(x) = \frac{x}{2} = \frac{1}{2}x$

$f'(x) = \frac{1}{2}$

$= 2\int \frac{1}{2} e^{\frac{x}{2}} \, dx - \int x^2 \, dx$

$= 2e^{\frac{x}{2}} - \frac{x^3}{3} + c$

$\frac{x}{2} = \frac{1}{2}x$

(b) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$

$f(x) = \sqrt{x} = x^{\frac{1}{2}}$

$f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$

$= \frac{1}{2\sqrt{x}}$

$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx = 2\int \frac{1}{2\sqrt{x}} \times e^{\sqrt{x}} \, dx$

$= 2e^{\sqrt{x}} + c$